Downstream Analysis

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RNA-seq Downstream Analysis

- After we got our DE gene list, we need to add biological meaning to this set of genes based on the following questions:

Which are the function of the modulated coding genes?

Is there any biological pathway affected by my treatment?

RNA-seq Downstream Analysis

- Then, we can run different analyses to get ideas about the function of the modulated genes:
  - Gene Ontology Mapping
  - Enrichment Analysis
  - Gene Set Enrichment analysis

Gene Ontology

The Gene Ontology (GO) describes our knowledge of the biological domain with respect to three aspects:

Molecular Function:
describe activities that occur at the molecular level, such as “catalysis” or “transport”.

Cellular Component:
Locations relative to cellular structures in which a gene product performs a function, either cellular compartments (e.g., mitochondrion), or stable macromolecular complexes of which they are parts (e.g., the ribosome).

Biological Process:
The larger processes, or ‘biological programs’ accomplished by multiple molecular activities. Examples of broad biological process terms are DNA repair or signal transduction.

They do not represent pathways

The GENEontology Consortium

GENEONTOLOGY Unifying Biology

About Ontology Annotations Downloads Help

ALLIANCE of GENOME RESOURCES FOUNDING MEMBER

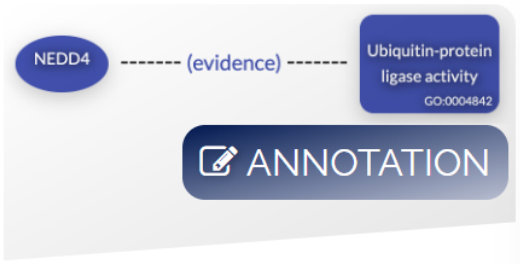
Any Ontology Gene Product

Hint: can use UniProt ID/AC, Gene Name, Gene Symbols, MOD IDs



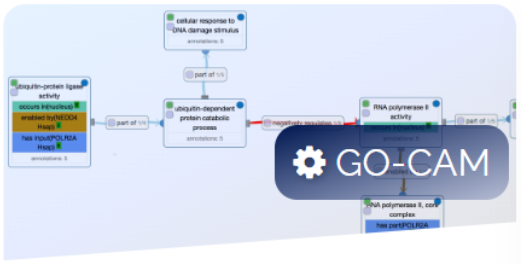
The network of biological classes describing the current best representation of the "universe" of biology: the molecular functions, cellular locations, and processes gene products may carry out.

GO Ontology Overview
Browse in AmiGO
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Statements, based on specific, traceable scientific evidence, asserting that a specific gene product is a real exemplar of a particular GO class.

GO Annotations Overview
Browse in AmiGO
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GO Causal Activity Model (GO-CAM) provides a structured framework to link standard GO annotations into a more complete model of a biological system.

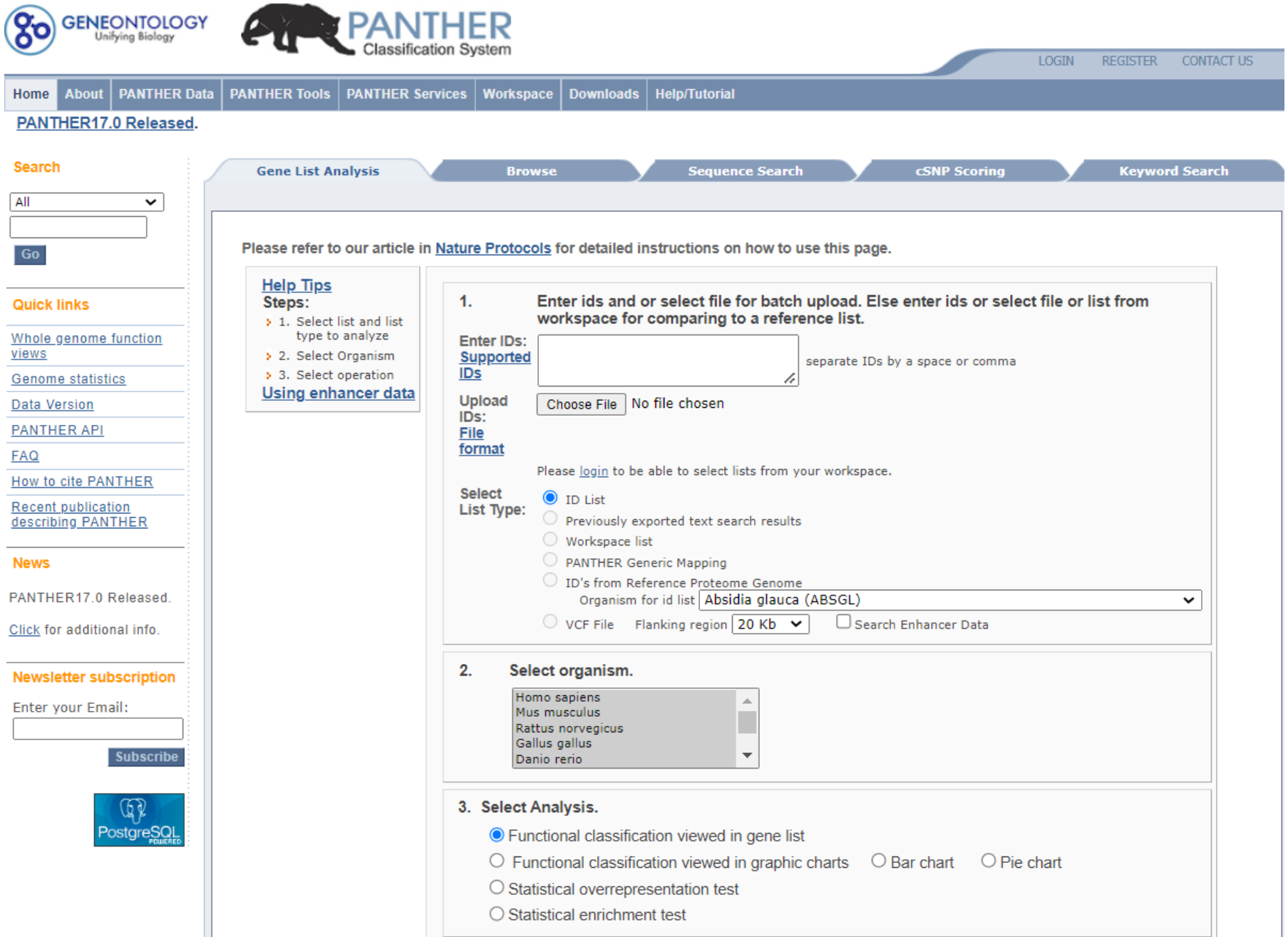
GO-CAM Overview
Browse GO-CAMs
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Tools to curate, browse, search, visualize and download both the ontology and annotations. Includes bioinformatic guides (Notebooks) and simple API access to integrate the GO into your research.

GO Tools Overview
GO APIs Guide
GO GitHub

Panther: a webtool for GO



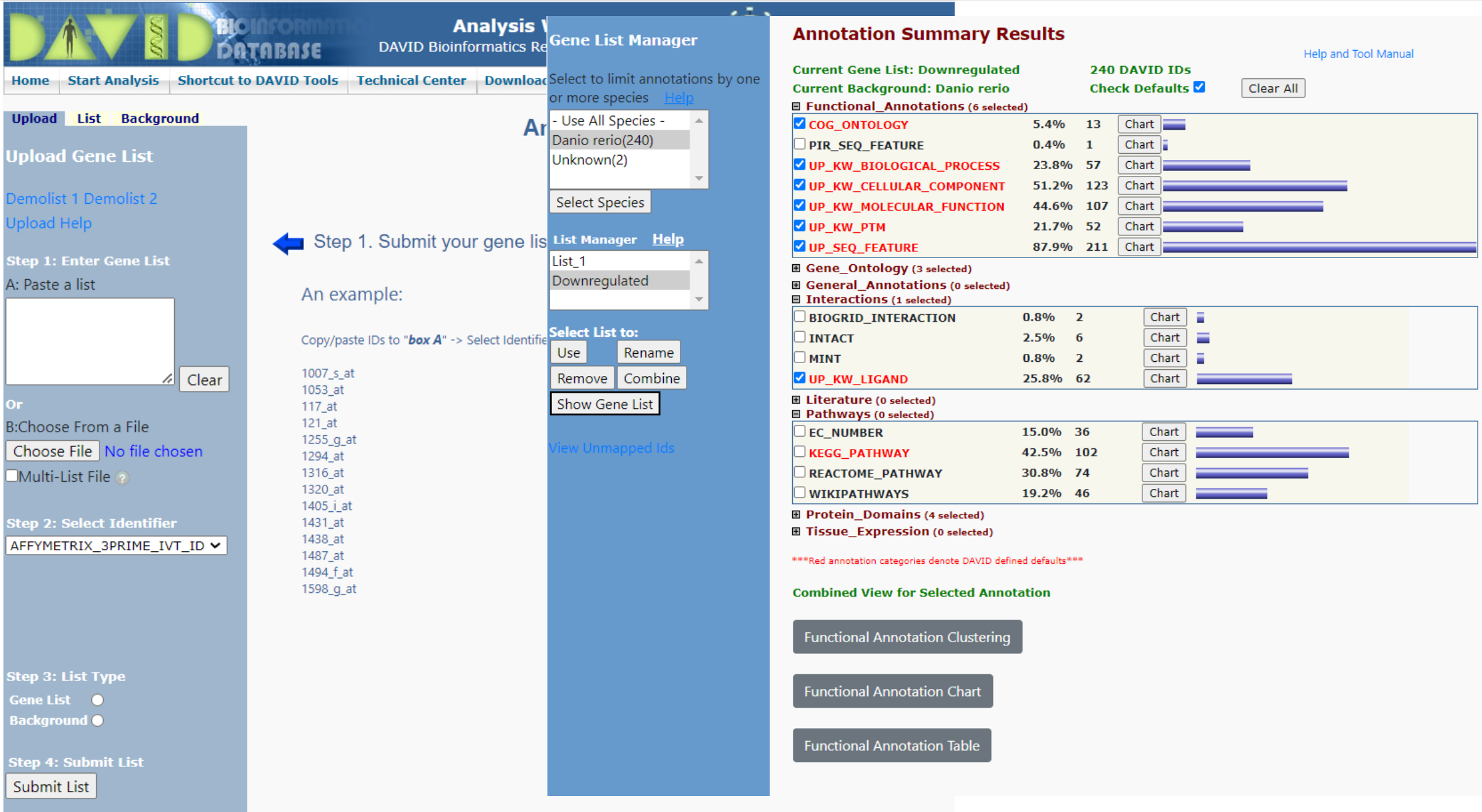
Enrichment Analysis

- It characterizes a gene list by looking at classes of genes representing functions that are overrepresented on the list and associated with your study

- The analysis with test statistically the overrepresentation of these gene classes and estimate if they are significant

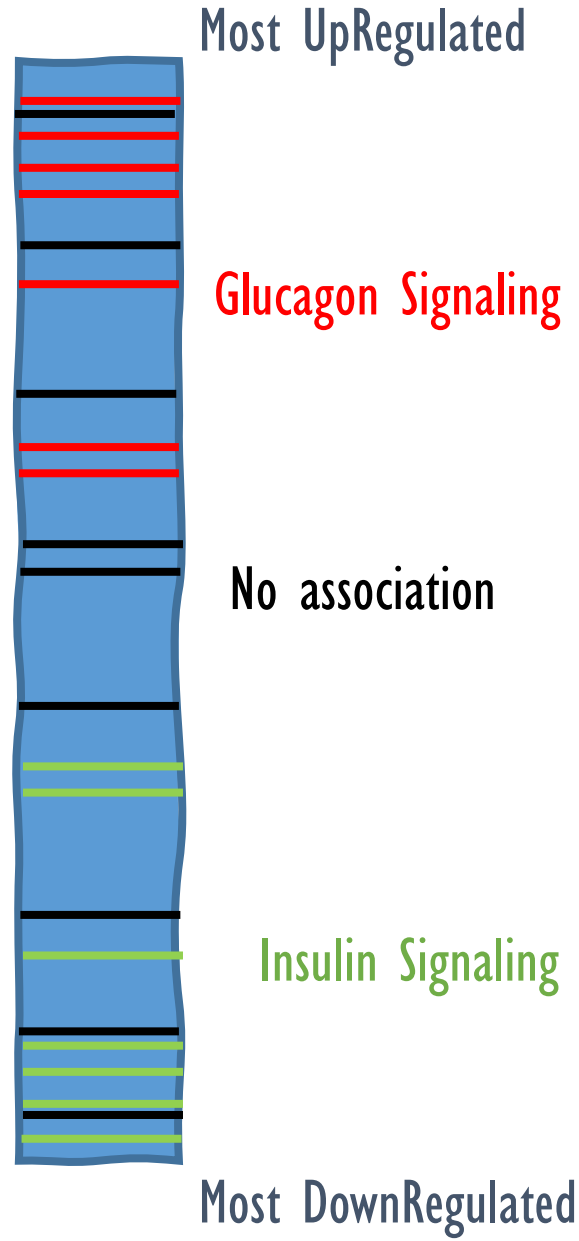
- For this analysis, the gene background used is essential. Your background must respond to the classes of genes used as input.
    - For transcriptome-wide modulated gene set the perfect background would be all the genes expressed in your data set.
    - For regulated kinases gene set a “kinome” background (all kinases annotated in the genome)

DAVID: Webtool for pathways analysis



GENE SET ENRICHMENT ANALYSIS (GSEA)

- Instead of comparing modulated genes list with a background list, we use a ranked list.
- This list will be organized in descending order based on the fold of change, p-value, etc
- Then, "functional terms" (GO, disease, etc) are mapped to the ranked list.
    - Genes upregulated that are enriched for a certain functional term will be at the top of the list
    - Genes Downregulated enriched for a certain term will be found at the bottom of the list
    - Terms not enriched will be mapped all over the list

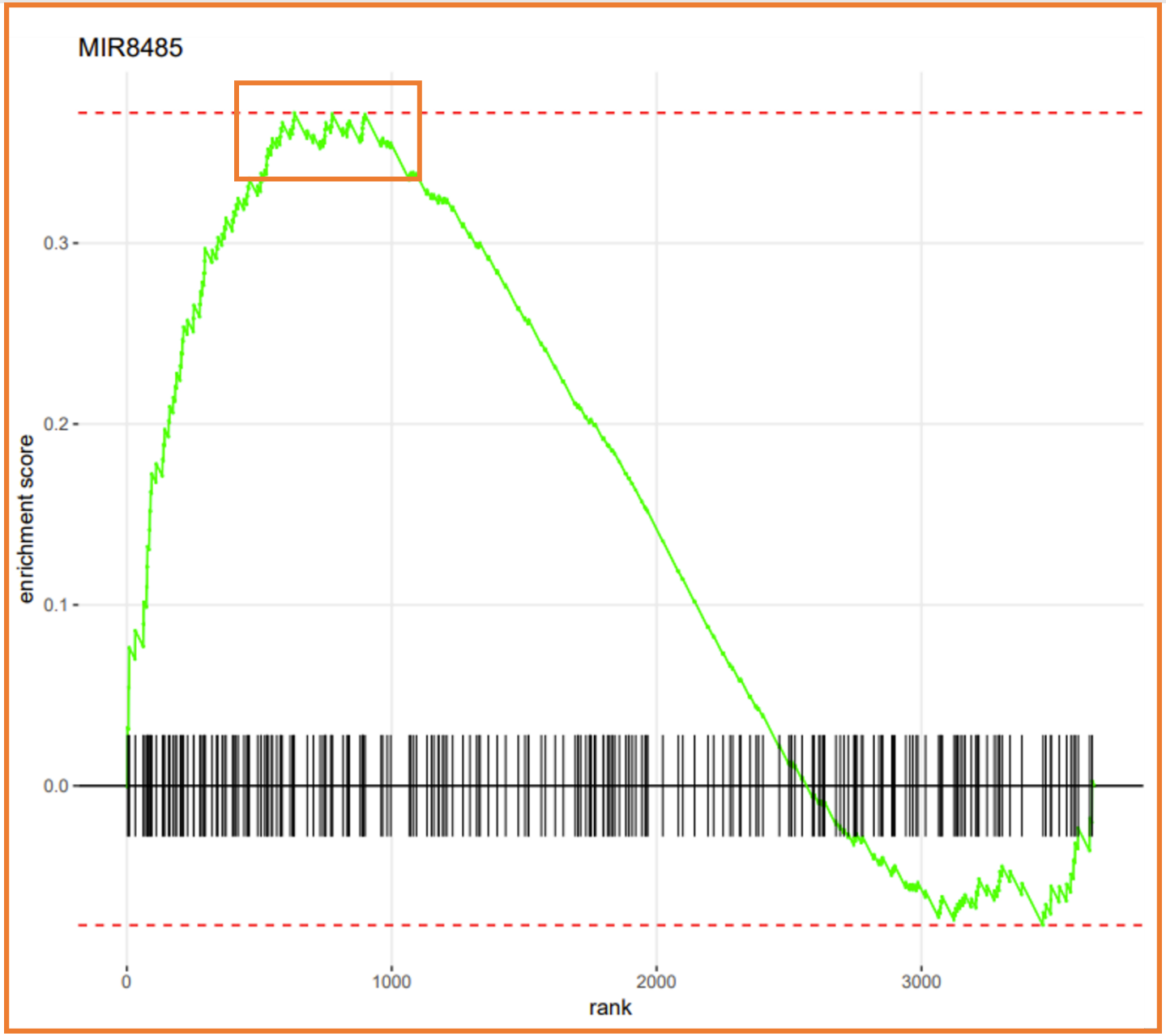


Interpreting Results from GSEA

Enrichment Score:

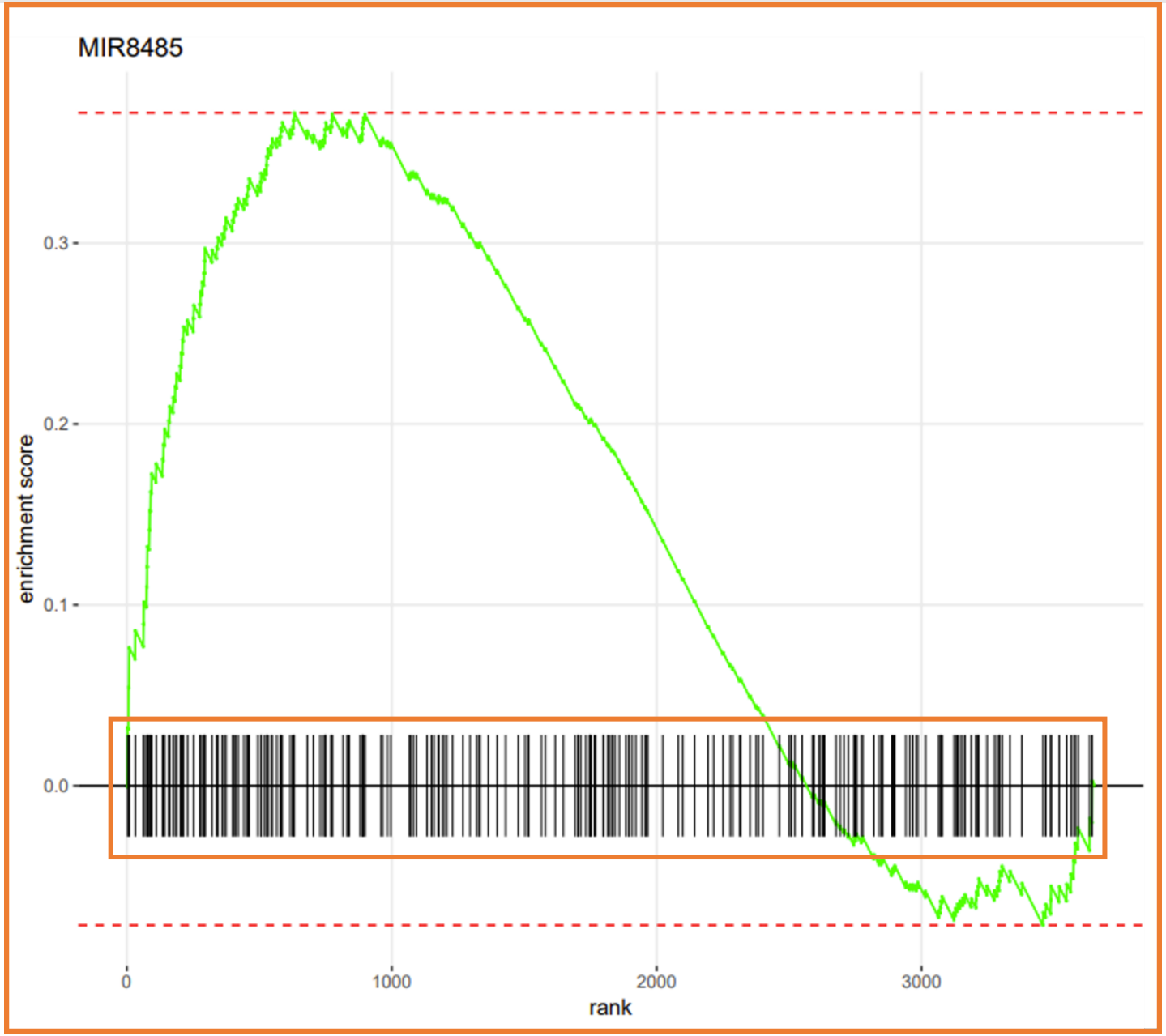
Which reflects the degree to which a gene set is overrepresented at the top or bottom of a ranked list of genes.

The score at the peak of the plot (the score furthest from 0.0) is the ES for the gene set. Gene sets with a distinct peak at the beginning (such as the one shown here) or end of the ranked list are generally the most interesting.



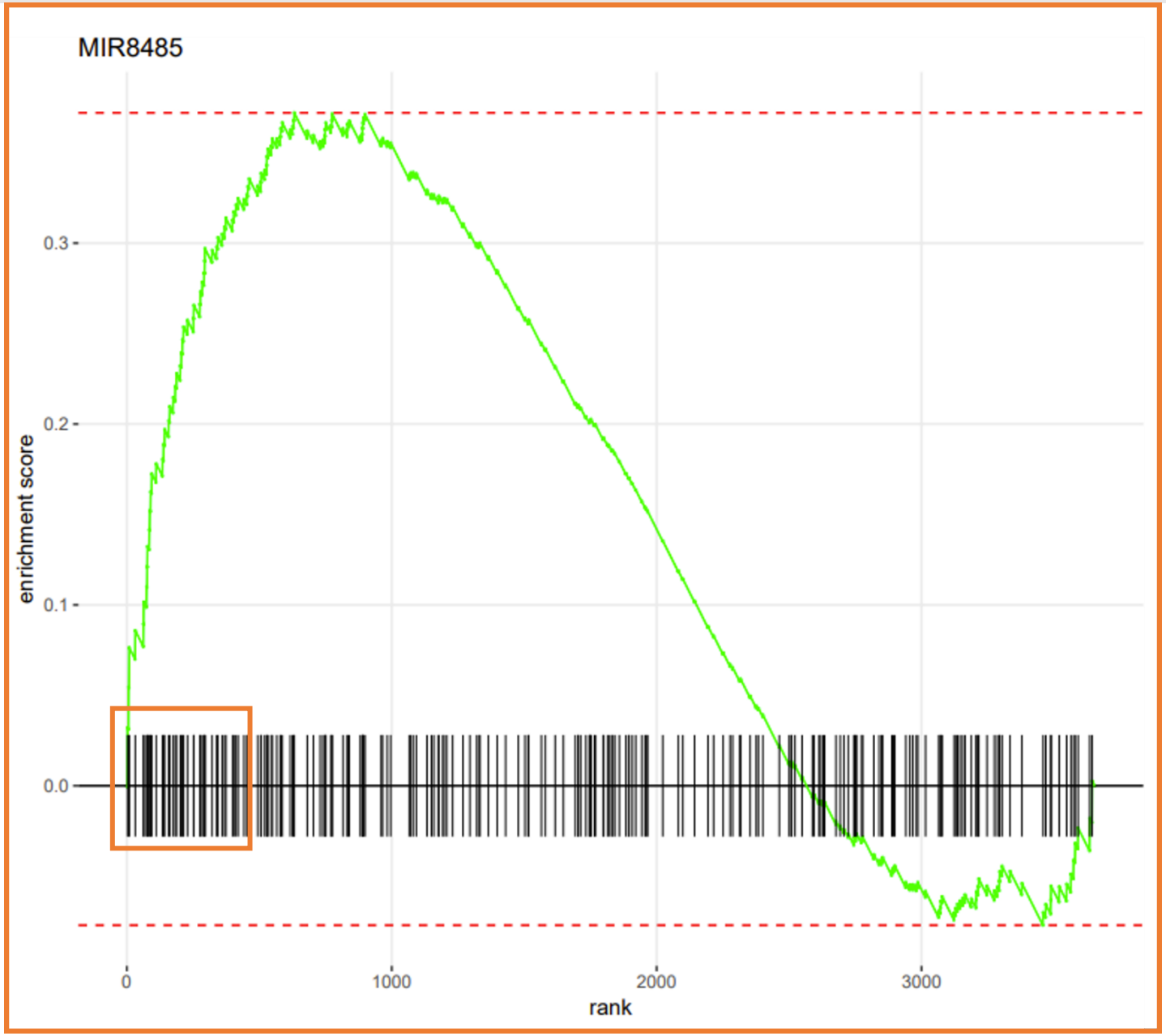
Interpreting Results from GSEA

Shows where the members of the gene list appears in the ranked list of genes



Interpreting Results from GSEA

Leading Edge Subset:
is the subset of members that contribute most to the ES. For a positive ES (such as the one shown here), the leading-edge subset is the set of members that appear in the ranked list prior to the peak score.



The end! … just coffee and then practical section

